

ABSTRACT

The problem of water pollution in ponds, lakes, marshes, and other forms of water bodies has gained significance because of waste materials, industrial discharges, and run-offs in urban areas. Pollution of water affects the aquatic life living in it, hence posing a threat to environmental balance; however, the traditional ways of cleaning water are costly and require a lot of energy, making it impossible for them to be used in large water bodies. Solar Powered Floating Hydroponic Rafting System is an innovative technology aimed at addressing the issue in question. It is a holistic approach for purifying water and cleaning air that involves hydroponics, phytoremediation, aeration, movement aid, internet of things, and solar energy. Hydroponic cultivation of aquatic plants enables absorption of toxins, unnecessary nutrients, and other contaminants found in the polluted water through the phytoremediation process. IoT sensors are responsible for monitoring the environment to assess the level of key parameters such as pH, TDS, temperature, humidity, and air quality. This process facilitates the constant observation of the environment in real time, thus improving analysis within the system. The system employs assisted mobility through thrusters, allowing the system to traverse different regions of the water body. Additionally, the system uses obstacles detection sensors to enhance navigational safety. This system offers an environmentally-friendly way of purifying and monitoring the environment based on low costs and renewable energy. This makes the system ideal in facilitating improved quality of the water in lakes and stagnant water bodies.

Contents

1	Introduction	12
1.1	Preamble	12
1.2	Motivation of the Project	13
1.3	Problem Statement	13
1.4	Block Diagram	14
1.5	Objectives	15
1.6	Organization of the Report.....	16
2	Literature Review with Gap in the Literature.....	17
2.1	Previous Research	17
2.2	Comparative Analysis of Existing Literature	21
2.3	Summary of Literature Review.....	23
3	Present Work Carried Out.....	25
3.1	Plant selection for phytoremediation.....	25
3.1.1	Hydroponic plant used	25
3.2	Hardware Requirements.....	27
3.3	Software Requirements	34
3.3.1	Embedded Programming Environment	34
3.3.2	IoT Platform (Blynk IoT)	34
3.3.3	Communication Protocol	35
3.3.4	Libraries Used.....	35
3.3.5	Data Visualization.....	35
3.3.6	Discussion on System Monitoring.....	36
3.4	System Design	36
3.4.1	Introduction.....	36

3.4.2	System Modeling	36
3.4.3	Proposed Design	37
3.4.4	Sensor Considerations	37
3.4.5	Propeller design	37
3.4.5.1	Propeller (Thruster) Design.....	37
3.4.5.2	Propeller Geometry and Design Features.....	38
3.4.5.3	CAD Model of the Propeller	38
3.4.5.4	Mathematical Modelling of Thrust Generation	39
3.4.5.5	Design Justification	39
3.4.6	Obstacle Detection and Reactive Avoidance.....	40
3.4.7	Reactive Control Logic	40
3.5	Mathematical Modeling	40
3.5.1	Water Purification Modelling.....	40
3.5.2	Buoyancy Analysis.....	41
3.6	Mathematical Design of Pump and Motor System	42
3.6.1	Water Pump Design.....	42
3.6.2	Aeration Requirement	43
3.6.3	Aeration System Implementation	44
3.6.4	Motor and Thruster Design	44
3.6.5	Motor Power Requirement	44
3.6.6	Control using PWM	45
3.7	Floating Mechanism.....	45
3.8	Power Analysis and Solar Energy System Design.....	46
3.8.1	System Power Requirement	46
3.8.2	Daily Energy Requirement.....	46
3.8.3	Battery Capacity Selection.....	46
3.8.4	Effective Load Adjustment:.....	47
3.8.5	Solar Panel Selection.....	47
3.8.6	Charging Time Estimation	48
3.8.7	System Feasibility	48
3.8.8	Design Justification	49
3.9	Water Flow and Purification Requirement	49

3.10	Environmental Parameter Requirements	49
3.11	IoT and Communication Requirements	50
3.12	System Integration	50
3.13	Design Considerations	50
3.14	Final Prototype Model	51
4	Results and Discussion.....	52
4.1	Experimental Proof	52
4.1.1	Experimental setup and Test conditions	52
4.1.1.1	Test Duration	52
4.1.1.2	Water Volume	52
4.1.1.3	Pollutant Source	52
4.1.1.4	Environmental Conditions.....	52
4.1.1.5	Sunlight Exposure	53
4.1.1.6	Sample Count.....	53
4.1.1.7	Data Collection Method	53
4.1.1.8	Test Objective.....	53
4.2	Validation of Objective 1: Water Purification using Hydroponics and Phytoremediation	55
4.2.1	Water quality validation (TDS reduction validation).....	55
4.2.2	Total Dissolved Solids (TDS).....	55
4.2.3	pH Analysis.....	56
4.3	Validation of Objective 2: Solar-Powered Aeration System	56
4.3.1	Air Quality Monitoring	56
4.3.2	Adaptive Aeration Control	57
4.3.3	Aeration Performance Observation	58
4.4	Validation of Objective 3: Assisted Mobility and Obstacle Detection.....	58
4.4.1	Hydrodynamic Drag Analysis.....	58
4.4.2	Thrust Analysis	60
4.4.3	PWM VS Velocity (EXPERIMENTAL MAPING)	60
4.4.4	Waterproof Durability Analysis	61
4.4.5	Obstacle Detection Performance Analysis	61

4.4.6	Detection Range	61
4.4.7	Measurement Accuracy	62
4.4.8	Response Time	62
4.4.9	Water Surface Reflection Effect	63
4.4.10	Performance Summary	63
4.5	Validation of Objective 4: IoT-Based Environmental Monitoring	63
4.5.1	Temperature Analysis.....	63
4.5.2	Air Quality and Water Quality Monitoring using IoT	64
4.5.3	Sensor Drift Analysis	65
4.5.4	Model Validation	65
4.5.5	Statistical Analysis and Repeatability Testing.....	66
4.5.5.1	Repeated Trials	67
4.5.5.2	Mean and Standard Deviation	67
4.5.5.3	Confidence Interval	67
4.5.5.4	Experimental Results with Statistical Analysis	67
4.5.5.5	Statistical Significance	68
4.6	System Performance Analysis.....	68
4.6.1	Power consumption validation	68
4.6.2	Power and Efficiency	68
4.6.3	Battery Endurance Analysis.....	68
4.7	Discussion.....	69
5	Conclusion	71
5.1	Advantages.....	71
5.2	Disadvantages	72
5.3	Future Scope	72
	REFERENCES	73

List of Figures

1.1	Block Diagram of the Proposed System	14
	Water Spinach (<i>Ipomoea aquatica</i>) used in the Hydroponic Phytoremediation System	
	27	
	Lightweight XPS Foam used for the Floating Raft Structure.....	27
	Net Pots used for Holding Hydroponic Plants	28
	ESP32-S3 Microcontroller used in the Proposed System	28
	4-in-1 pH Sensor Module used for Water Quality Monitoring.....	29
	Analog TDS Sensor Module Used for Water Quality Measurement.....	29
	DS18B20 Water Temperature Sensor Used in the Proposed System	30
	MQ-135 Gas Sensor Used for Air Quality Monitoring.....	30
	DHT11 Temperature and Humidity Sensor Used in the Proposed System	31
	Three-Blade Thruster Propeller Used for Water Propulsion	31
	IRLZ44N MOSFET Module Used for PWM-Based Switching Control	32
	12V, 12W Solar Panel Used in the Proposed System	32
	12V, 5Ah Rechargeable Battery Used in the Proposed System	33
	Ultrasonic Sensor Module Used for Obstacle Detection	33
	Connectors and Wiring Used in the Proposed System	34
	Designed 80 mm three-blade propeller showing (a) top view, (b) bottom view, (c) isometric view, and (d) side view	38
4.1	Experimental validation of the proposed system showing (a) deployment of the floating raft in water, (b) system stability during operation	53
4.2	Experimental validation of the proposed system showing real-time monitoring of water and air quality parameters using the Blynk IoT interface.....	54
4.3	Water Quality Monitoring Output Displayed by the Proposed System.....	55
4.4	Air Quality Monitoring Output Displayed by the Proposed System.....	57

4.5	Adaptive aeration control system where pump speed is dynamically adjusted based on real-time average values of pH, TDS, and water temperature	58
4.6	Drag force variation with velocity	59
4.7	Thrust variation with PWM input.....	60
4.8	Acceleration profile of the raft for varying PWM inputs	61
4.9	System performance under continuous water exposure	62
4.10	Air Quality, Water Quality and Temperature Monitoring using the Blynk IoT Platform	65
4.11	Variation in sensor readings over time	66
4.12	Comparison of Theoretical and Experimental Results	66
4.13	Battery voltage variation over time during operation.....	69

List of Tables

2.1	Comparison of Existing Literature	21
3.1	Power consumption of system components	46
4.1	Validation of Water Purification Model	55
4.2	Air Quality Sensor Validation	57
4.3	Validation of Hydrodynamic Drag Model	59
4.4	Ultrasonic Sensor Detection Range	62
4.5	Statistical Analysis of TDS Measurements	67
4.6	Power Consumption Analysis	68

CHAPTER 1

INTRODUCTION

Pollution in the environment has been identified as one of the significant concerns in recent years, especially in lakes and other natural water bodies. Human activities such as industrial discharge, sewage dumping, and other forms of pollution have contaminated water resources, thereby indirectly affecting aquatic life.[1]. Moreover, deterioration in air quality has emerged as another serious concern that not only impacts the ecosystem but also human health [2]. To solve these issues, we are proposing a Solar Powered Floating Garden model for cleaning air and water. This model works based on the concept of hydroponic, which means cultivation of plants without soil along with phytoremediation, the process through which pollutants and toxic compounds can be extracted from water with the help of plants [3]. Our proposed solution uses solar energy as well as real-time monitoring through IoT-enabled devices. The objective of this project is to develop an eco-friendly and sustainable mechanism to ensure the quality of water and air in the environment [4].

1.1 PREAMBLE

As stated earlier, the Solar Powered Floating Garden System proposed in this essay will be an innovation that will be able to provide a sustainable and reliable solution for enhancing water and air quality through various methods. For instance, the Solar Powered Floating Garden System will incorporate the principles of hydroponics, phytoremediation, aeration, mobility, monitoring through IoT technology, and energy generation from solar energy. Hydroponics involves growing plants in water without using any other medium except for nutrients absorbed in the water. Through phytoremediation, aquatic plants will absorb contaminants, excessive nutrients, and toxic elements from the water body [5]. In addition to purifying water, the plants will be able to enhance the quality of air in the environment.

One key feature of the proposed Solar Powered Floating Garden System is the incorporation of mobility, purification, and monitoring. As compared to most of the floating treatment facilities that do not incorporate mobility into their design, the platform will be able to move around across different parts of a water body using thrusters [6]. With such mobility, the process

of interacting with water for purification purposes will be much improved.

Continuous monitoring of parameters such as pH, Total Dissolved Solids (TDS), temperature, humidity, and air quality will be undertaken using IoT-based sensors [7]. Such monitoring will allow real-time observations of the environmental parameters. Controlled aeration is another aspect of the Solar Powered Floating Garden System in order to increase the dissolved oxygen levels in water[8]. Controlled water circulation will also enhance the process of purifying water in order to maintain proper conditions for growth of the plants. Remote-controlled navigation and obstacle detection will ensure safe movements across lakes and water bodies. Solar energy will enable the use of such equipment outdoors.

1.2 MOTIVATION OF THE PROJECT

Effluents from sewage discharge, industrial waste, and urban runoff are increasingly contaminating lakes and other water bodies. Water pollution affects aquatic life, damages the environment, and can also lead to serious health-related problems. Traditional water purification techniques are often costly, energy-intensive, and difficult to implement effectively in lakes and stagnant water bodies. In addition, only a limited number of systems provide continuous environmental monitoring and purification coverage.

The motivation behind the proposed Solar Powered Floating Garden System is to come up with a sustainable eco-friendly solution to better water and air quality. Hydroponics, phytoremediation, aeration and assisted mobility, IoT sensor monitoring and solar power are all part of the same floating platform. Hydroponically grown plants can absorb excessive pollutants and nutrients from polluted water, and the real-time monitoring and controlled water movement can increase purification efficiency and environmental observation.

1.3 PROBLEM STATEMENT

Urban lakes are increasingly affected by pollution due to sewage, industry and urban run-off, ponds, swamps, wetlands and other slow flowing and water stagnant water bodies, which affect water quality, air quality, aquatic life and the environment. It is not suitable for continuous purification in such water bodies because most of the existing purification systems are very expensive and unmovable, and have limited environmental monitoring. In consequence, a continuous water and air purification, a real-time monitoring, assisted mobility, and a long-term

outdoor operation system supported by renewable energy sources is needed, which is sustainable and eco-friendly.

1.4 BLOCK DIAGRAM

The block diagram of the proposed Solar Powered Floating Hydroponic Rafting System is displayed in Figure 1.1. It shows a general view of the architecture and communication between sensing, control, power management, communication, propulsion and purification modules of the system.

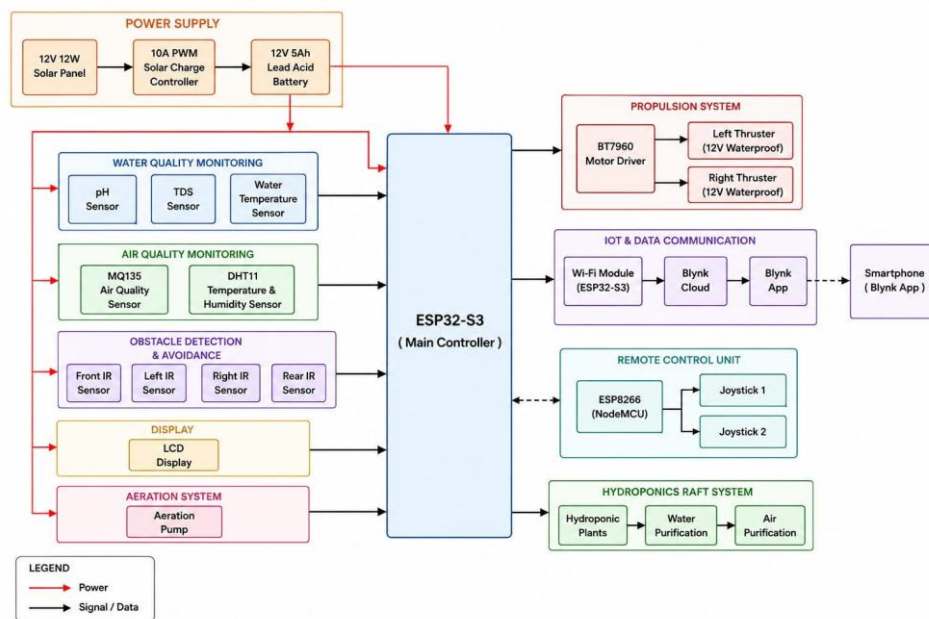


Figure 1.1: Block Diagram of the Proposed System

The proposed system is composed of several interrelated modules in order to enable environmental monitoring, water purification and floating independently. Using ESP32-S3 Microcontroller as the core control unit, it processes the data it collects. These convey data from a variety of sensors and control modules. The power supply module comes with 12V, 12W solar panel, solar charge controller, and a 12V, 5Ah rechargeable battery. This is a subsystem that furnishes renewable and perpetual power to provide power to the sensors, pumps, motors and communication modules. The water quality monitoring module is made up of pH, TDS and DS18B20 temperature. Real-time water quality monitoring sensors of water quality parameters. The air quality monitoring module consists of the MQ135 gas sensor and DHT11 sensor. Measuring atmospheric conditions (air quality, temperature and humidity) around the floating platform.

The obstacle detection and avoidance module is created with the help of IR sensors which are put around the raft. To identify obstacles in the area and help navigate safely. The propulsion system includes Move forward, reverse, left and right using waterproof thrusters controlled by the BTS7960 motor driver. Movement of floating raft to the left and right. Plants of water spinach are introduced to system for phytoremediation and natural purification. The IoT and communication module allows wireless data transfer via Wi-Fi. With the ESP32-S3 and Blynk cloud platform. Environmental data can be monitored and it will be available in real time via a smartphone application. A module with remote control using ESP8266 Hand control of the floating system is possible with and can be fitted with joystick units when necessary. Overall, the integrated system offers an intelligent, efficient and environmentally friendly solution. The system can be used for water purification, environmental monitoring and sustainable lake management.

1.5 OBJECTIVES

The main objectives of the proposed Solar Powered Floating Garden System are as follows:

- To design floating garden system for natural water purification, healthy plant growth and removal of pollutants from the lake water to integrate hydroponics and phytoremediation.
- To utilize solar energy for operating an aeration system that improves dissolved oxygen levels and supports aquatic life.
- To establish a platform that will float and be movable with assistance and detect obstacles to move safely across water bodies.
- To deploy the sensors based on IoT for continuous water quality monitoring and environmental parameters.

1.6 ORGANIZATION OF THE REPORT

This report is divided into five chapters, with each chapter explaining a different part of the proposed Solar Powered Floating Hydroponic Rafting System.

Chapter 1: Introduction gives an overview of the project and explains the environmental issues related to water pollution and poor environmental conditions. It discusses the need for sustainable purification methods and introduces the proposed system along with its objectives, block diagram, and overall project overview.

Chapter 2: Literature Review with Gap in the Literature explains the existing methods and technologies used for water purification, hydroponics, phytoremediation, IoT-based monitoring, and solar-powered systems. The chapter also compares the methodologies used in previous research works and identifies the limitations of the existing systems. Based on this analysis, the research gap addressed by the proposed system is presented.

Chapter 3: Present Work Carried Out describes the design and development of the proposed system. It includes the hardware components, floating hydroponic structure, plant selection, solar-powered aeration system, obstacle detection mechanism, remote-control operation, and IoT-based monitoring system. This chapter also includes the mathematical modelling and power analysis of the system.

Chapter 4: Results and Discussion presents the experimental observations and performance analysis of the system. The chapter discusses the water quality monitoring results, air quality analysis, purification process, power consumption, and overall working performance of the floating purification system.

Chapter 5: Conclusion and Future Scope tells the outcomes of the project and highlights the advantages of the proposed system. It also discusses the limitations of the current implementation and suggests possible future improvements such as advanced monitoring systems, autonomous operation, and large-scale deployment for environmental management applications.

LITERATURE REVIEW WITH GAP IN THE LITERATURE

2.1 PREVIOUS RESEARCH

The proposed solution of the research deals with development of an intelligent hydroponics farming system using Deep Water Culture (DWC) and Nutrient Film Technique (NFT). IoT devices including sensors, Raspberry Pi and machine learning algorithms can be applied in the analysis of temperature, humidity, water conditions, and nutrients requirement. Machine learning contributes to automation of nutrient management, plant growth and development, as well as reduces water utilization. The authors emphasize that use of IoT and machine learning in the hydroponic farming system enables improvement of the process due to continuous monitoring and control. Nevertheless, the paper addresses indoor and smart agriculture rather than environmental purification or floating phytoremediation systems [9].

The proposed study involves a hydroponic farming technology which utilizes artificial intelligence along with IoT technologies for automated farming. Sensors help to monitor temperature, humidity, water level, and nutrients. Machine learning algorithms are employed in predicting the nutrient requirements and detecting the conditions of plants' health. The use of IoT technologies contributes to enhancing hydroponic farming because of reduced water consumption and continuous monitoring. Nevertheless, the focus of the research remains at indoor hydroponic farming and smart agriculture, not environmental purification or floating treatment systems [10].

This paper discusses a monitoring system for hydroponic plant cultivation based on IoT technology. Various types of sensors are used to analyze pH, electrical conductivity, temperature, humidity, nutrient concentration, and water levels. The collected data is then analyzed via Blynk IoT platform and a mobile app. It can be concluded that the IoT technology will significantly increase the efficiency of hydroponic farming via constant monitoring and automated environmental control. Nonetheless, the presented study is oriented mostly at indoor hydroponic farming and smart gardening systems and does not touch upon such topics as water purification and phytoremediation systems [11].

This research paper presents the idea of an automated organic hydroponic farming system which is based on using vermicompost tea as an organic fertilizer for cultivating lettuces. In this research paper, the sensors were used for monitoring pH, electrical conductivity, temperature, humidity, and light intensity in the hydroponic system. Furthermore, the research was conducted using IoT technology. It may be concluded that automated farming can significantly help in growing crops while reducing the use of chemical fertilizers. Nevertheless, the focus of this paper is limited by hydroponics and organic farming [12].

This study explores the reuse of domestic wastewater treated through different methods, such as aeration, disinfection, and addition of nutrients, for cultivating plants using a hydroponic farm. Various parameters in treated wastewater such as pH value, dissolved oxygen, nitrate concentration, and heavy metal presence are measured in order to enhance their usability for plant growth. This research proves that wastewater reuse in hydroponics could decrease freshwater use while maintaining sustainability in agricultural products. However, this research primarily focuses on wastewater reuse in hydroponics rather than on environmental monitoring or automation in purifying wastewater [13].

This research is based on the development of an automated hydroponic farm driven by solar energy and IoT technology. Solar energy serves as an energy source for solar panels and batteries used in this farming method; IoT devices such as sensors, NodeMCU ESP8266, LED lights, and water pumps are used to maintain proper conditions for plants' growth. Sensors measure temperature, humidity, and light intensity, thus helping to perform various farming tasks automatically. This research proves that solar energy, IoT, and data analytics can be utilized effectively in hydroponic farming operations. However, the study mainly focuses on indoor farming rather than water purification [14].

This paper focuses on the design of solar powered hydroponic farm system with IoT technology to support smart agriculture. It uses different sensors, solar panels, batteries, and automated monitoring methods to create optimal conditions for plant growth. This study indicates that renewable energy sources along with IoT can be helpful in efficient and effective farm operations by decreasing reliance on power generated from fossil fuels. However, the research mostly concentrates on hydroponics farming rather than water purification using IoT technology or floating phytoremediation systems [15].

In this study, the focus is on the use of phytoremediation which is an environmentally friendly

technique for cleaning contaminated soil and water bodies. There is information about various types of phytoremediation techniques like phytoextraction, rhizofiltration, phytostabilization, and phytovolatilization which help to remove contaminants. The paper also provides information about the absorption capacity of some plants for absorbing toxic metals like lead, cadmium, and chromium from contaminated places. The study indicates that phytoremediation is a cost-effective and sustainable solution for purifying soil, water, and air. However, the emphasis of the paper is not on the use of IoT and automation technology in monitoring or purification processes [16]. Floating treatment wetlands used for pollutants removal from stormwater via plants grown in rafts are under examination in the paper. Plants like *Carex riparia*, *Carex pseudocyperus*, and *Phalaris arundinacea* are considered in relation to metal absorption capacity. It has been proven that floating treatment wetlands allow effective removal of heavy metals including cadmium, copper, lead, and zinc from polluted stormwater. The research indicates that plant growth and root development have significant impact on removal rate. Nonetheless, the research primarily aims at pollution remediation rather than smart monitoring or automation of floating treatment [17].

The study considers the application of floating wetland islands for water purification and sustainable crop growth. Floating wetlands combine biological features associated with floating plant communities, microbial activity, and the design of floating platforms aimed at improvement of water conditions naturally. In the context of the study, it is indicated that proper choice of plants, appropriate design of floating platform and optimal water flow management have an impact on increasing of efficiency of the process. However, the primary purpose of the research was floating wetland island creation rather than IoT-implementation [18].

The study is about the use of floating aquatic plants for removing heavy metals from contaminated water. Techniques such as phytoextraction, rhizofiltration, phytostabilization, and phytovolatilization are discussed for pollutant removal. Floating plants such as duckweed, water lettuce, water hyacinth, and *Salvinia* are highlighted for their ability to absorb toxic metals from water. The study shows that floating aquatic plants can effectively remove metals such as cadmium, lead, copper, and chromium through root absorption. However, the work mainly focuses on metal removal and large-scale water treatment rather than environmental monitoring or automated purification systems [19].

The study is about Floating Treatment Wetlands (FTWs) used for water and wastewater

treatment. FTWs use floating platforms with aquatic plants and microbial activity to improve water purification naturally. The review highlights factors such as plant selection, hydraulic retention time, and floating structure design for improving treatment efficiency. The study explains that FTWs are low-cost and environmentally friendly systems suitable for stormwater treatment, wastewater purification, and nutrient removal. However, the work mainly focuses on wetland treatment design rather than IoT-based monitoring or automated floating systems [20]. The study discusses artificial floating islands for wastewater treatment in urban areas. Aquatic plants, such as *Eichhornia crassipes*, *Pistia stratiotes*, and *Salvinia molesta*, are used for pollutant absorption and nutrient removal. Parameters like pH, turbidity, dissolved oxygen, nitrate, and phosphate concentration are measured for the evaluation of treatment effectiveness. According to the study, the application of floating islands results in successful nutrient pollution reduction and improved wastewater quality using biological purification processes. The research pays primary attention to plant-based wastewater treatment and does not cover smart monitoring or automation processes [21].

The paper analyzes a hydroponic system for sewage treatment by water hyacinths. The study examines the removal of pollutants like COD, BOD, phosphates, ammoniacal nitrogen, and suspended solids in the wastewater. As a result, the research demonstrates that water hyacinth has efficient nutrient absorption properties and can be used to increase dissolved oxygen concentrations in water. The research explains the process of hydroponics' contribution to pollutant elimination in a small-sized sustainable environment. However, the paper is focused on wastewater treatment without any IoT-based monitoring or automated environmental assessment processes [22].

In this study, the treatment of sewage water through water hyacinth for removing contaminants from the sewage water has been analyzed. In this study, parameters like COD, BOD, phosphates, ammoniacal nitrogen, and suspended solids have been taken into account. The study shows that the combination of water hyacinth along with microbes improves the absorption of contaminants and the purification process. The study explains other aspects as well such as harvesting and hydraulic retention time. However, the study mainly emphasizes sewage water treatment instead of any monitoring and automation technologies [23].

In this study, the topic covered is Floating Treatment Wetlands (FTWs). It is used for purification of wastewater and environmental rehabilitation. The review includes floating

platforms, aquatic plants, aeration system, and microbial activities for efficient treatment of wastewater. It is observed that FTWs are environment-friendly and cost-effective techniques that can be used for treatment of sewage water, industrial wastewater, and stormwater purification. The role of interaction of plant root with microorganism is crucial in nutrient and contaminant removal. However, the study is mainly focused on treatment system instead of monitoring or automation [24].

In this work, the utilization of vetiver grass for Floating Treatment Wetlands (FTW) is explored. Various types of substrates like leaf litter, pumice stones, corn cob, and mixed substrates are investigated to examine the role of the substrate on plant development and pollution treatment effectiveness. Variables including dissolved oxygen, total dissolved solids, and plant growth are measured in the process of water purification. It can be pointed out that the selection of substrates greatly influences the effectiveness of plant survival and pollutant removal through FTWs. Nonetheless, the paper mainly concerns itself with river purification and substrate effectiveness rather than IoT-based monitoring or automated floating systems [25].

2.2 COMPARATIVE ANALYSIS OF EXISTING LITERATURE

Table 2.1 presents a comparative analysis of the reviewed literature based on methodology, technologies used, application area, and identified limitations.

Table 2.1: Comparison of Existing Literature

Ref.	Methodology / System	Technologies Used	Application Area	Limitation Identified
[9]	Smart hydroponic farming using DWC and NFT	IoT, Raspberry Pi, Machine Learning	Indoor hydroponic farming	Limited focus on environmental purification and floating systems
[10]	AI-based hydroponic monitoring system	AI, IoT, ANN, KNN, Sensors	Smart agriculture	No phytoremediation or floating treatment application
[11]	IoT-based hydroponic cultivation monitoring	Blynk IoT, Sensors, Mobile Monitoring	Indoor gardening	Limited focus on wastewater purification

Ref.	Methodology / System	Technologies Used	Application Area	Limitation Identified
[12]	Automated or-organic hydroponics system	IoT, Sensors, LED lighting	Organic crop cultivation	Does not address floating purification systems
[13]	Wastewater reuse in hydroponic cultivation	Floating hydroponics, Water quality analysis	Wastewater reuse	No IoT monitoring or automation
[14]	Solar-powered indoor hydroponics	Solar panel, IoT analytics, ESP8266	Smart indoor farming	Limited environmental remediation focus
[15]	IoT-based plant growth monitoring	Cloud monitoring, Hydroponics	Smart agriculture	Does not focus on water purification
[16]	Phytoremediation for pollutant removal	Hyperaccumulator plants, Rhizofiltration	Environmental remediation	No automation or IoT integration
[17]	Floating treatment wetlands for metal removal	Wetland plants, Floating rafts	Stormwater treatment	No smart monitoring system
[18]	Floating wetland islands review	Floating hydroponics, Phytoremediation	Water treatment and crop production	Limited automation and monitoring
[19]	Floating aquatic plants for heavy metal removal	Aquatic plants, Phytoremediation	Heavy metal remediation	No IoT-based analysis
[20]	Floating Treatment Wetlands review	FTWs, Biofilms, Macrophytes	Wastewater treatment	Focused mainly on wetland design
[21]	Artificial floating islands for wastewater treatment	Aquatic macrophytes, Phytoremediation	Urban wastewater treatment	Lack of automated monitoring
[22]	Water hyacinth hydroponic sewage treatment	Hydroponics, Phytoremediation	Sewage treatment	No smart monitoring or automation

Ref.	Methodology / System	Technologies Used	Application Area	Limitation Identified
[24]	FTW design and implementation review	Floating platforms, Aeration systems	Wastewater purification	Limited IoT integration
[25]	Vetiver grass based FTW system	Floating wetlands, Substrate analysis	Urban river remediation	No automated or IoT-based control

2.3 SUMMARY OF LITERATURE REVIEW

Based on the literature review conducted, it is found that there are several studies carried out by various researchers in the fields of hydroponics, phytoremediation, floating treatment wetlands, monitoring with IoT, and solar powered environmental systems. Several studies and research related to smart hydroponic farming with Internet of Things (IoT) and Artificial Intelligence (AI) with machine learning and sensor-based monitoring to enhance plant growth and water use efficiency. Other studies focused on using aquatic plants like water hyacinth, duckweed, vetiver grass and *Salvinia* for the removal of pollutants, nutrients and heavy metals from polluted water by phytoremediation and floating wetland systems.

The reviewed studies further showed that hydroponic culture and aquatic plants play a pivotal role in water quality enhancement by employing natural purification processes. To monitor hydroponic farming conditions in real-time, some research works applied to IoT platforms, cloud monitoring and automated environmental control systems. Other studies examined floating treatment wetlands, artificial floating islands and hydroponic wastewater treatment system to achieve a sustainable environmental remediation and wastewater purification.

Based on the comparative analysis, it was noticed that most of the systems discussed are limited to individual applications like smart agriculture, wastewater treatment, phytoremediation or floating wetlands, independently. There are numerous studies available that offer effective purification methods, but fail to include sensor-enabled environment, mobility support, automation, and/or integration with renewable energy sources. Likewise, there are a few hydroponic systems for IoT which are primarily associated with indoor cultivation, and controlled agriculture, not with cleaning up the environment.

The comparison table also indicates that only a few systems integrate hydroponics, floating

phytoremediation, solar energy, aeration, real-time monitoring, assisted mobility and environmental analysis in one single integrated and compact system. The proposed Solar Powered Floating Hydroponic Rafting System uses hydroponic phytoremediation, floating treatment mechanisms, assisted mobility, aeration, solar powered, IoT monitoring and air-water quality analysis, as a single solution for continuous environmental purification and monitoring.

CHAPTER 3

PRESENT WORK CARRIED OUT

This chapter presents the design and implementation of the smart floating system developed for water and air purification. It covers the key aspects of the project, including the hardware components, software modules, overall system architecture, and the control logic that governs its operation. The system brings together sensors, IoT-based communication, solar energy, and phytoremediation methods into a single working model. By integrating these elements, the prototype is designed to effectively monitor environmental conditions and contribute to the improvement of both water and air quality.

3.1 PLANT SELECTION FOR PHYTOREMEDIATION

3.1.1 Hydroponic plant used

The selection of water spinach (*Ipomoea aquatica*) in the suggested floating hydroponic phytoremediation system is based on the fact that the plant grows quickly, has good adaptability to aquatic conditions, and possesses high absorption properties. Water spinach can be used in hydroponic and phytoremediation processes due to the plant's property of growing in water without the presence of soil. The plant's vast fibrous root structure ensures close contact with water containing pollutants and is ideal for the process of nutrients and pollutant uptake and water cleansing.

The primary rationale behind the choice of water spinach is that it can remove excessive nutrients like nitrogen and phosphorus. These elements cause pollution of water bodies through the process known as eutrophication. Moreover, water spinach will help lower the level of Total Dissolved Solids (TDS), suspended particulate matter, and some other organic pollutants due to nutrient removal and bacteria activity in the vicinity of roots.

The suitability of water spinach for use in float hydroponics is also very high because of its ability to adapt well to living in aquatic environments with a high content of nutrients and continue growing there. As opposed to most aquatic plants, water spinach does not require much care and grows rather fast. It is capable of growing up to 30-100 cm long plants with hollow

stems and long green leaves, whereas the roots develop rather fast and have a high surface area used for purifying purposes.

The second significant reason why water spinach can be used in hydroponic systems lies in its ability to effectively use nutrients in these conditions. Water spinach grows normally when provided with macro-elements including nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulfur (S) as well as micro-elements like iron (Fe), zinc (Zn), manganese (Mn), copper (Cu), and boron (B). Additionally, the root system enhances oxygen transfer due to increased water circulation around the roots. This not only ensures a healthy aquatic environment but also increases purification efficiency. Moreover, the growth of this aquatic plant occurs at a relatively quick rate and the plant has an adaptive mechanism which makes it suitable for use in various environments and even outdoors in different water bodies such as ponds and lakes.

The ability of water spinach to grow rapidly compared to other ornamental plants used in the treatment process, as well as its efficiency in terms of absorbing pollutants and growing rapidly, makes it ideal for use in such a system. Additionally, the plant can be obtained from cheap sources without any difficulty.

The factors considered while selecting the plant species included the efficiency of pollutant absorption, fast growth rate, low maintenance requirement, and adaptation to hydroponic conditions. Taking into account all these factors, the water spinach plant is ideal for the Solar Powered Floating Hydroponic Rafting System.



Figure 3.1: Water Spinach (*Ipomoea aquatica*) used in the Hydroponic Phytoremediation System

3.2 HARDWARE REQUIREMENTS

The proposed system integrates various hardware components to enable real-time monitoring, control, and purification of water and air. It combines sensor modules, a microcontroller-based control unit, actuation devices, and a solar-powered energy system. These components work together to collect environmental data, process it using the microcontroller, and perform automated actions such as aeration. The major hardware components used in the system are described below:

1. Hydroponic Raft (Floating Platform)

The hydroponic raft is made of lightweight XPS foam with dimensions of 35 cm × 55 cm and acts as the base structure of the system. It supports plant growth and holds all components while floating on water. The raft ensures buoyancy and stability, allowing the system to operate efficiently in lakes. It also contains holes for placing net pots with plants.



Figure 3.2: Lightweight XPS Foam used for the Floating Raft Structure

2. Net Pots to hold Plants

Net pots are used for holding plants like water spinach within the hydroponics set-up. The plant roots come into direct contact with the water; thus, they absorb the required nutrients as well as toxins. Such plants have a key part in phytoremediation whereby they eliminate pollutants in water.



Figure 3.3: Net Pots used for Holding Hydroponic Plants

3. ESP32-S3 Microcontroller

The ESP32-S3 is the controlling device in the system that processes information from sensors and handles the communication process. Information from all the sensors is collected by the controlling device, and it uploads the data into the cloud platform through Wi-Fi. The microcontroller makes it possible to monitor and automate activities.



Figure 3.4: ESP32-S3 Microcontroller used in the Proposed System

4. pH Sensor

The use of the pH probe will allow one to monitor in real-time the acidity or basicity of

the water. This information is necessary for the estimation of water quality and determining if the environment is suitable for growing plants. The importance of keeping a proper pH value cannot be underestimated because it directly influences the process of nutrient absorption by plants.



Figure 3.5: 4-in-1 pH Sensor Module used for Water Quality Monitoring

5. TDS Sensor (Total Dissolved Solids)

TDS Sensor is used for calculating the total amount of dissolved matter contained within the water sample. This gives information about the impurities and purity of the water sample in question. It is important to note that higher levels of TDS indicate the presence of a greater amount of dissolved impurities in the water. On the other hand, a low reading indicates clean water.

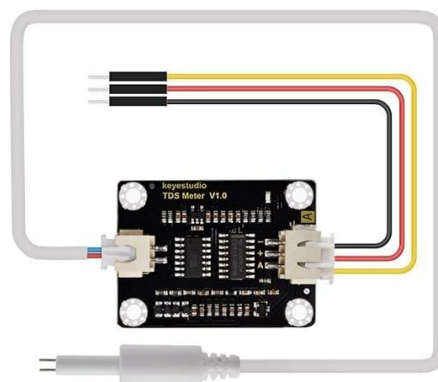


Figure 3.6: Analog TDS Sensor Module Used for Water Quality Measurement

6. Water Temperature Sensor (DS18B20)

The water temperature sensor constantly measures the temperature of the water inside the system. It is very essential to maintain an appropriate temperature as it will help the plants to grow well and aid in the biological and chemical reactions that occur during water purification.



Figure 3.7: DS18B20 Water Temperature Sensor Used in the Proposed System

7. MQ-135 Gas Sensor

The gas sensors MQ-135 are used for testing the quality of the environment's air by detecting the various gases, such as CO₂, NH₃, smoke, benzene, among others, in the environment. This helps test the environmental condition near the floating hydroponics for continuous air quality monitoring. The analog signals from this sensor depend on the amount of gases found in the environment, which is analyzed using the microcontroller, ESP32-S3. It is possible to find out how much effect the water body has on the air quality from this sensor.



Figure 3.8: MQ-135 Gas Sensor Used for Air Quality Monitoring

8. DHT11 Humidity and Temperature Sensor

The DHT11 sensor is used for sensing the environmental conditions of temperature and humidity around the system. Monitoring these environmental factors is necessary because

these factors influence the performance of the plant growth in hydroponic phytoremediation. The maintenance of optimal environmental conditions enables better functioning of the system. The output from these sensors is digital information that is processed by the ESP32-S3 microcontroller.

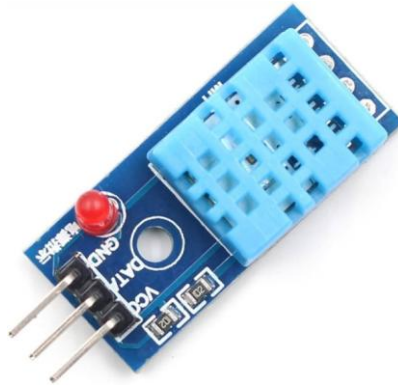


Figure 3.9: DHT11 Temperature and Humidity Sensor Used in the Proposed System

9. Underwater Thrusters (Motors)

The thrusters drive the platform into movement over the water surface and make it possible for the platform to move to other places apart from remaining stationary. Not only do the thrusters ensure mobility but also enable the circulation of water in order to enhance nutrient uptake by the plant roots. The constant movement prevents water stagnation near the platform, thus enabling the maintenance of higher oxygen content in the water.



Figure 3.10: Three-Blade Thruster Propeller Used for Water Propulsion

9. IRLZ44N MOSFET Module

This IRLZ44N MOSFET is applied as a switching component in the system. The proposed system applies the MOSFET in controlling the high current component by means of the PWM signals produced by the ESP32-S3 microcontroller. Although its function is comparable to a relay, it presents benefits like fast switching, low power dissipation, and effective performance. Hence, it is applicable for the control of devices such as motors and aerators. By controlling the motor and aerator through PWM, the motor speed and aeration intensity may be regulated according to the needs of the system.



Figure 3.11: IRLZ44N MOSFET Module Used for PWM-Based Switching Control

11. Solar Panel (12V, 12W)

The renewable energy source that is recommended to be used with the proposed floating hydroponics is the solar panel having 12 volts and 12 watts capacity. Solar panel is capable of converting energy from sunlight into electric energy to recharge the battery. This energy will be responsible for running the entire system continuously. This will serve as an energy source for the system's components, including ESP32-S3, sensors, aeration system, and motors.



Figure 3.12: 12V, 12W Solar Panel Used in the Proposed System

12. Battery (12V, 5Ah Rechargeable Battery) The 12V, 5Ah rechargeable battery acts as an energy storage device for the electrical power produced by the solar panel. It helps ensure that the system keeps running when there is not enough light during the daytime or any at all during nighttime. The energy stored is used to supply the ESP32-S3 microcontroller, sensors, aeration system, and motors, ensuring smooth and uninterrupted operation of the floating hydroponics system.

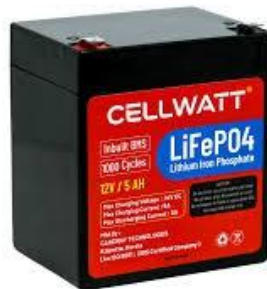


Figure 3.13: 12V, 5Ah Rechargeable Battery Used in the Proposed System

10. Ultrasonic sensor

Ultrasonic sensor is used to detect obstacles and calculate the distance of the floating platform from obstacles. The sensor is used by transmitting ultrasonic waves and detecting their reflection back from the surfaces around it. By detecting obstacles in the environment when in motion, it is helpful for safe movement through water bodies. Ultrasonic sensors are used widely in many fields due to their ease of operation and dependable operation.



Figure 3.14: Ultrasonic Sensor Module Used for Obstacle Detection

13. Wiring and Connectors Wires and connectors are used to connect all components in

the system. They ensure proper transmission of power and signals. Good quality wiring reduces energy loss and improves safety. Proper connections are essential for system performance.



Figure 3.15: Connectors and Wiring Used in the Proposed System

3.3 SOFTWARE REQUIREMENTS

This Proposed system includes a number of software tools and platforms that are used to collect, process, visualize, and wirelessly transmit information. With the aid of these software tools, the hydroponic raft system will be able to operate in real-time mode.

3.3.1 Embedded Programming Environment

The ESP32-S3 microcontroller is programmed using the Arduino IDE. The Arduino platform provides a user-friendly environment for developing, compiling, and uploading code to the microcontroller.

- Arduino IDE
- ESP32 Board Support Package
- Embedded C/C++ programming language

3.3.2 IoT Platform (Blynk IoT)

IoT Platform (Blynk IoT) The IoT platform of Blynk is utilized to monitor and visualize the sensor data via a mobile application. The wireless connection is established between the ESP32-S3 microcontroller and the user interface. The following parameters are displayed in the Blynk application:

- pH value of water
- Total Dissolved Solids (TDS)
- Water temperature
- Ambient temperature
- Humidity

The ESP32-S3 transmits sensor data over Wi-Fi to the Blynk cloud, which is then displayed on the mobile dashboard using virtual pins.

3.3.3 Communication Protocol

Wi-Fi is used as the communication medium between the ESP32-S3 and the Blynk platform. The system uses HTTP-based communication through the Blynk library for data transmission.

3.3.4 Libraries Used

The following libraries are used in the software implementation:

- WiFi.h – For wireless connectivity
- BlynkSimpleEsp32.h – For Blynk IoT integration
- Wire.h – For I2C communication (LCD)
- LiquidCrystal_I2C.h – For LCD display
- DHT.h – For temperature and humidity sensing
- HardwareSerial.h – For UART communication (pH sensor)

3.3.5 Data Visualization

The sensor data is displayed in two ways:

- Locally on an LCD screen for on-site monitoring
- Remotely on the Blynk mobile application for real-time access

3.3.6 Discussion on System Monitoring

The integration of embedded programming with IoT platforms enables efficient real-time monitoring of environmental parameters. The use of Blynk IoT provides a user-friendly interface for data visualization, making the system accessible and scalable for real-world applications.

3.4 SYSTEM DESIGN

The system requirements describe the functions and operations necessary for the suggested solar-powered floating hydroponic system. The system utilizes sensors, controls, energy management, and water purifications systems.

3.4.1 Introduction

The system is engineered to continuously check the water and air quality levels while conducting the process of purification by making use of hydroponics. The system functions autonomously by making use of solar energy.

3.4.2 System Modeling

Modeling of the system is done considering interactions between various subsystems and functional modules:

- **Sensing Module:** Consists of pH, TDS, DS18B20, MQ135, and DHT11 sensors used for collecting environmental information.
- **Control Module:** Responsible for ESP32-S3 that collects sensor values and carries out commands for pumps and communication.
- **Actuation Module:** Water pump and aerator carry out purification of water using circulation and oxidation.
- **Power Management Module:** Utilizes solar panel, charge controller, and battery for continuous energy source.

3.4.3 Proposed Design

This design comprises four layers which are integrated into one system:

- Input Layer : Sensors capture real-time data.
- Processing Layer : ESP32-S3 processor analyzes and processes the data.
- Output Layer : Pumps and aerators execute tasks.
- Communication Layer : Internet of things application monitors system performance.

The architecture guarantees automation, energy savings, and scalability.

3.4.4 Sensor Considerations

Correct selection of sensors plays an important role in ensuring precise monitoring:

- pH Sensors : Used to measure water acidity or alkalinity
- TDS Sensors : Used to detect pollution levels of dissolved solids in water
- DS18B20 : Used to accurately measure water temperatures
- MQ135 : Used to detect air pollution and gases
- DHT11 : Used to measure temperature and humidity in the air

3.4.5 Propeller design

3.4.5.1 Propeller (Thruster) Design

Thrust force on the hydroponics floating raft is provided through the custom-made three-blade propeller. The three-blade propeller will provide a sufficient thrust force for the movement of the raft while consuming less power.

A diameter of 80mm and three blades are incorporated in the designed propeller to be used in providing thrust force on the raft. Three blades have a curved shape to make them more efficient in displacing water.

Having three blades is ideal because there is a balance of both thrust force and power consumption. Two blades consume less power but cause vibration when being used. More blades consume more power.

The size of the propeller is smaller and will not affect buoyancy when being mounted.

3.4.5.2 Propeller Geometry and Design Features

- Diameter: 80 mm
- Number of blades: 3
- Blade type: Curved profile
- Material: Lightweight polymer (3D printed)
- Application: Low-speed water propulsion

3.4.5.3 CAD Model of the Propeller

Below is the figure showing the CAD model of the propeller from various angles. Top and bottom views show symmetry in the blades. Isometric view depicts the curve in the blades. Side view denotes that the blades have an incline angle and pitch to provide thrust through acceleration of water backwards.



Figure 3.16: Designed 80 mm three-blade propeller showing (a) top view, (b) bottom view, (c) isometric view, and (d) side view.

3.4.5.4 Mathematical Modelling of Thrust Generation

The thrust generated by the propeller is based on momentum theory and is given by:

$$T = \dot{m}(V_{exit} - V_{inlet})$$

where:

- T = Thrust force (N)
- $\dot{m}$ = Mass flow rate of water (kg/s)
- V_{exit} = Velocity of water leaving the propeller (m/s)
- V_{inlet} = Velocity of incoming water (m/s)

The power required by the propeller is given by:

$$P = T \cdot V$$

where:

- P = Power (W)
- V = Velocity of the raft (m/s)

3.4.5.5 Design Justification

It was engineered in such a way that it would offer maximum efficiency when used at low speed. It was done by adjusting the curvatures of the blades and their pitch angles to ensure an optimal balance between force generation and drag force. Three blades make it easy for the raft to generate effective propulsion and reduce vibrations that would result in smooth operation of the raft. In addition, it is small in size and causes minimal disturbance to the water around it. With this propeller set up with the power supply, the raft will be easily maneuverable and facilitate exposure of the polluted water to the roots of the hydroponic plants.

3.4.6 Obstacle Detection and Reactive Avoidance

The developed system does not implement autonomous navigation or path planning algorithms such as SLAM or trajectory optimization. Instead, it employs a simple assisted obstacle detection and reactive avoidance mechanism.

An ultrasonic sensor is used to detect nearby obstacles in the forward direction. When an obstacle is detected within a predefined threshold distance, the system responds by stopping or altering the motion of the thrusters. This reactive approach ensures basic collision avoidance while maintaining manual or remote control of the raft.

This method is computationally simple and suitable for low-power embedded systems, although it does not provide full autonomy or intelligent navigation capabilities.

3.4.7 Reactive Control Logic

The obstacle avoidance mechanism follows a threshold-based control logic:

- If no obstacle is detected: Normal operation
- If obstacle is detected: Motor speed is reduced or stopped
- Directional adjustment based on manual input

This approach ensures safe operation without requiring complex algorithms, making it suitable for real-time embedded applications.

3.5 MATHEMATICAL MODELING

A study on the efficiency of purification, stability, and energy expenditure of the floating hydroponics purification method has been done through mathematical modeling. The mathematical model plays a critical role in establishing the efficiency of the process

3.5.1 Water Purification Modelling

The pollutant removal efficiency of the phytoremediation system is calculated using:

$$\eta = \frac{C_i - C_f}{C_i} \times 100 \quad (3.1)$$

where:

- C_i = Initial pollutant concentration
- C_f = Final pollutant concentration
- η = Pollutant removal efficiency (%)

The pollutant absorption rate by the hydroponic plants is expressed as:

$$R = \frac{(C_i - C_f)V}{t} \quad (3.2)$$

where:

- R = Pollutant removal rate
- V = Volume of water treated
- t = Treatment time

The hydroponic plants absorb excess nutrients, nitrogen compounds, phosphates, Total Dissolved Solids (TDS), and organic pollutants from the water, thereby improving water quality through phytoremediation.

3.5.2 Buoyancy Analysis

The floating raft remains stable due to the buoyant force acting on the XPS foam structure. According to Archimedes' principle, the buoyant force is given by:

$$F_b = \rho g V \quad (3.3)$$

where:

- F_b = Buoyant force (N)

- ρ = Density of water (kg/m^3)
- g = Acceleration due to gravity (9.81 m/s^2)
- V = Volume of displaced water (m^3)

The condition for stable floating is:

$$F_b \geq W \quad (3.4)$$

where:

- W = Total weight of the floating raft and components

The lightweight XPS foam structure with dimensions of $35 \text{ cm} \times 55 \text{ cm}$ provides sufficient buoyancy to support the hydroponic plants, sensors, solar panel, battery, and electronic modules without sinking.

3.6 MATHEMATICAL DESIGN OF PUMP AND MOTOR SYSTEM

3.6.1 Water Pump Design

The water pump is chosen to provide adequate flow within the root zone of the hydroponics system. The required rate of flow is determined according to the quantity of water that needs to be circulated.

The flow rate of the pump is:

$$Q = \frac{V}{t}$$

where,

- Q = Flow rate (m^3/s)
- V = Volume of water (m^3)
- t = circulation time (s)

For a small-scale raft system, assuming a water volume of approximately 0.05 m³ and a circulation time of 60 seconds:

$$Q = \frac{0.05}{60} = 8.33 \times 10^{-4} \text{ m}^3/\text{s}$$

The hydraulic power required by the pump is given by:

$$P = \rho g Q H$$

where,

- ρ = Density of water (1000 kg/m³)
- g = Acceleration due to gravity (9.81 m/s²)
- H = Head (m)

Assuming a head of 0.5 m:

$$P = 1000 \times 9.81 \times 8.33 \times 10^{-4} \times 0.5 \approx 4.08 \text{ W}$$

This confirms that a low-power DC pump is sufficient for the system.

3.6.2 Aeration Requirement

The role of the aerator pump in the increase of oxygen concentration in the water body is performed. The oxygen transfer rate is affected by the air flow rate and contact area. A simplified relationship can be presented as follows:

$$OTR \propto Q_{air}$$

where Q_{air} represents the air flow rate.

3.6.3 Aeration System Implementation

An air pump was integrated into the floating platform to enhance dissolved oxygen (DO) levels in the water. The pump operates using solar-powered energy through a 12V battery system. Continuous bubbling action ensures oxygen transfer from air to water, supporting aquatic life and improving water quality.

3.6.4 Motor and Thruster Design

The thrust generated by the motor is required to overcome the hydrodynamic drag acting on the raft.

The drag force is given by:

$$F_d = \frac{1}{2} \rho C_d A v^2$$

Under steady-state conditions, the thrust force balances the drag force:

$$T = F_d$$

Assuming:

- $C_d = 0.9$
- $A = 0.05 \text{ m}^2$
- $v = 1 \text{ m/s}$

$$F_d = \frac{1}{2} \times 1000 \times 0.9 \times 0.05 \times (1)^2 = 22.5 \text{ N}$$

Thus, the motor must generate a thrust of approximately 22.5 N to maintain motion.

3.6.5 Motor Power Requirement

The mechanical power required is:

$$P = F \times v$$

$$P = 22.5 \times 1 = 22.5 \text{ W}$$

Considering efficiency losses:

$$P_{input} = \frac{P}{\eta}$$

Assuming efficiency $\eta = 0.7$:

$$P_{input} \approx \frac{22.5}{0.7} \approx 32 \text{ W}$$

This justifies the selection of a DC motor capable of delivering approximately 30–35 W power.

3.6.6 Control using PWM

The motor speed is controlled using PWM signals. The average voltage applied to the motor is given by:

$$V_{avg} = D \times V_s$$

where,

- D = Duty cycle
- V_s = Supply voltage

By varying the duty cycle, the motor speed and thrust can be controlled effectively.

3.7 FLOATING MECHANISM

Stability and buoyancy will be ensured by the design of the floating structure:

- Made from lightweight water-resistant material (foam/PVC)
- Provides proper balance in weight distribution
- Allows hydroponics to grow plants
- Prevents tilting and sinking

3.8 POWER ANALYSIS AND SOLAR ENERGY SYSTEM DESIGN

3.8.1 System Power Requirement

The total power consumption of the system is estimated by summing the power requirements of individual components. The main loads include two DC thruster motors, a water circulation pump, an aeration pump, and sensing/control electronics.

Table 3.1: Power consumption of system components

Component	Voltage (V)	Current (A)	Power (W)
DC Motor (each)	12	1.0	12
Two Motors	12	2.0	24
Water Pump	12	0.4	4.8
Air Pump	12	0.25	3.0
Sensors + ESP32 + Drivers	5/12	0.15	1.8
Total	—	—	33.6 W

To account for wiring, driver losses, and conversion inefficiencies, a safety factor of 15% is included:

$$P_{\text{total}} = 33.6 \times 1.15 \approx 38.6 \text{ W}$$

3.8.2 Daily Energy Requirement

Assuming the system operates for $t = 2$ hours per day:

$$E_{\text{daily}} = P_{\text{total}} \times t = 38.6 \times 2 = 77.2 \text{ Wh}$$

3.8.3 Battery Capacity Selection

A 12 V, 5 Ah sealed lead-acid (SLA) battery is selected.

$$E_{\text{battery}} = V \times Ah = 12 \times 5 = 60 \text{ Wh}$$

However, for SLA batteries, only about 70% of the capacity is usable (to avoid deep discharge):

$$E_{\text{usable}} = 0.7 \times 60 = 42 \text{ Wh}$$

Observation:

$$E_{\text{usable}}(42 \text{ Wh}) < E_{\text{daily}}(77.2 \text{ Wh})$$

Thus, the battery alone cannot support continuous 2-hour operation at full load. In practice, the motors do not operate at maximum power continuously, and the system operates under partial load conditions.

3.8.4 Effective Load Adjustment:

Assuming average operating load is 60%:

$$P_{\text{avg}} = 0.6 \times 38.6 \approx 23.2 \text{ W}$$

$$E_{\text{daily,avg}} = 23.2 \times 2 = 46.4 \text{ Wh}$$

Now,

$$E_{\text{usable}}(42 \text{ Wh}) \approx E_{\text{daily,avg}}(46.4 \text{ Wh})$$

Hence, the selected battery is suitable for short-duration operation with partial loading.

3.8.5 Solar Panel Selection

A 12 V, 12 W solar panel is used for charging.

The daily energy generation is given by:

$$E_{\text{solar}} = P_{\text{panel}} \times H$$

where H is the effective sunlight duration (assumed 5 hours/day):

$$E_{\text{solar}} = 12 \times 5 = 60 \text{ Wh}$$
 Considering

80% system efficiency:

$$E_{\text{solar,eff}} = 0.8 \times 60 = 48 \text{ Wh}$$

3.8.6 Charging Time Estimation

Battery charging current:

$$I = \frac{P}{V} = \frac{12}{12} = 1 \text{ A}$$

Charging time:

$$t_{\text{charge}} = \frac{Ah}{I} = \frac{5}{1} = 5 \text{ hours}$$

Including losses:

$$t_{\text{charge,real}} \approx 6 \text{ to } 7 \text{ hours}$$

3.8.7 System Feasibility

The feasibility of the proposed system was evaluated based on the estimated daily energy consumption and solar power generation capacity.

- Daily energy consumption (average): $\approx 46.4 \text{ Wh}$
- Solar energy generated: $\approx 48 \text{ Wh}$

Conclusion:

The selected 12 V, 5 Ah battery and 12 W solar panel combination is sufficient for low-power, short-duration operation. The solar panel is capable of replenishing the energy consumed during daily operation under favorable sunlight conditions.

3.8.8 Design Justification

The system is designed to operate under energy-constrained conditions with intermittent motor usage. The selected battery provides adequate backup, while the solar panel ensures sustainable operation by recharging the battery during daytime.

This configuration achieves a balance between cost, weight, and energy efficiency, making it suitable for small-scale floating purification systems.

3.9 WATER FLOW AND PURIFICATION REQUIREMENT

Effective purification requires continuous water circulation:

- Ensures nutrient absorption by plant roots
- Pump maintains water movement
- Aerator increases dissolved oxygen
- Enhances biological purification efficiency

3.10 ENVIRONMENTAL PARAMETER REQUIREMENTS

The following environmental parameters were considered for maintaining suitable conditions for hydroponic plant growth and water purification.

- pH Range: 6.5 – 8.5
- TDS: < 500 ppm
- Temperature: 20°C – 30°C
- Humidity: 40% – 70%

3.11 IOT AND COMMUNICATION REQUIREMENTS

The proposed system requires IoT-based communication for real-time monitoring and environmental data transmission.

- Wi-Fi communication using NodeMCU
- Real-time data transmission
- Cloud-based monitoring
- Mobile app visualization

3.12 SYSTEM INTEGRATION

The different hardware and monitoring components are integrated together for proper system operation and environmental monitoring.

- Sensors collect environmental data
- NodeMCU processes and analyzes inputs
- Relay controls pumps and aerators
- Solar system powers entire setup
- IoT platform displays real-time data

3.13 DESIGN CONSIDERATIONS

The following design considerations were taken into account during the development of the proposed system.

- Low power requirement for solar-powered operation
- Continuous flow of water for purification
- Stability of the floating platform
- Energy efficiency and conservation
- Real-time data acquisition and analysis

3.14 FINAL PROTOTYPE MODEL

The complete design incorporates all these features to create one integrated system:

- Sensors measure water and air quality
- NodeMCU processes and sends the data
- Water pumps carry out circulation and aeration
- Solar energy drives the system
- Smartphone application shows real-time information about the system

The entire design is sustainable and economically feasible for water purification.

RESULTS AND DISCUSSION

4.1 EXPERIMENTAL PROOF

4.1.1 Experimental setup and Test conditions

The developed floating hydroponic system was experimentally tested under controlled and real-time environmental conditions to evaluate its performance.

4.1.1.1 *Test Duration*

The system was operated continuously for a duration of approximately 2 hours per trial. Multiple trials were conducted over different days to ensure consistency in results.

4.1.1.2 *Water Volume*

The experiments were carried out in a water body with an approximate volume of 80–100 liters. This ensured sufficient space for the floating raft to operate freely while maintaining measurable pollutant concentration.

4.1.1.3 *Pollutant Source*

To simulate contaminated water conditions, natural stagnant water containing organic impurities, suspended particles, and dissolved solids was used. This represents real-world polluted water scenarios.

4.1.1.4 *Environmental Conditions*

The testing was performed under the following conditions:

- Ambient Temperature: 30–34°C
- Water Temperature: 25–28°C
- Relative Humidity: 30–40%

- Air Quality: Moderate (based on MQ135 readings)

4.1.1.5 Sunlight Exposure

The system was tested under natural sunlight conditions with an average exposure of 4–5 hours per day. This ensured effective operation of the solar panel and battery charging system.

4.1.1.6 Sample Count

A total of 5 experimental trials were conducted. Sensor readings such as pH, TDS, temperature, and air quality were recorded at regular intervals during each trial.

4.1.1.7 Data Collection Method

Sensor data was collected using the ESP32-based monitoring system and transmitted to the Blynk IoT platform for real-time visualization and storage. Manual observations were also recorded for validation.

4.1.1.8 Test Objective

The primary objective of the experiment was to evaluate:

- Water quality improvement (pH, TDS, water temperature)
- Air quality monitoring performance
- Stability of the floating platform
- Power sustainability using solar energy



Figure 4.1: Experimental validation of the proposed system showing (a) deployment of the floating raft in water, (b) system stability during operation



Figure 4.2: Experimental validation of the proposed system showing real-time monitoring of water and air quality parameters using the Blynk IoT interface.

The system was subjected to test experiments in order to analyze its working in real-time conditions. Water samples were taken over a period of time, and parameters like TDS and pH were analyzed. There was a consistent decrease in the pollution levels observed, showing the efficiency of the phytoremediation process. The process of water circulation and aeration through the plants helped achieve this. The propulsion mechanism was also experimented upon in similar manner with varying levels of PWM inputs and recording the movement of the raft as a consequence. Thus, the results have shown that the system works fine with control input.

The experiments were carried out in a water body with an approximate volume of 80–100 liters. This ensured sufficient space for the floating raft to operate freely while maintaining measurable pollutant concentration.

These findings prove that the system is feasible enough in practical life.

4.2 VALIDATION OF OBJECTIVE 1: WATER PURIFICATION USING HYDROPONICS AND PHYTOREMEDIATION

4.2.1 Water quality validation (TDS reduction validation)

To simulate contaminated water conditions, natural stagnant water containing organic impurities, suspended particles, and dissolved solids was used. This represents real-world polluted water scenarios.



Figure 4.3: Water Quality Monitoring Output Displayed by the Proposed System

The theoretical and experimentally measured TDS values obtained during the purification process are compared in Table 4.1.

Table 4.1: Validation of Water Purification Model

Time (days)	Theoretical TDS (ppm)	Measured TDS (ppm)	Error (%)
0	500	500	0
5	430	445	3.5
10	370	390	5.4
15	320	340	6.2

The decreasing trend in TDS confirms the effectiveness of phytoremediation and aligns with the exponential decay model.

4.2.2 Total Dissolved Solids (TDS)

Total Dissolved Solids (TDS) refer to the amount of dissolved solids in water. TDS is a critical parameter in determining the quality of water and its nutrient content. The total dissolved solids

value is determined from the voltage output of the sensor through the empirical relation

$$TDS = (133.42V^3 - 255.86V^2 + 857.39V) \times 0.5 \quad (4.1)$$

where V is the compensated voltage.

The observed TDS values ranged between 200 ppm and 500 ppm, indicating moderate nutrient concentration suitable for hydroponic applications.

4.2.3 pH Analysis

The pH value indicates the acidity or alkalinity of water and is measured on a logarithmic scale ranging from 0 to 14.

$$pH = -\log_{10}[H^+] \quad (4.2)$$

where $[H^+]$ represents the concentration of hydrogen ions in the solution.

The recorded pH levels in the experimental system were noted to fall within the 6.5 to 7.5 range, which denotes neutral acidity. The specified range is ideal for plant growth in the hydroponics set-up.

4.3 VALIDATION OF OBJECTIVE 2: SOLAR-POWERED AERATION SYSTEM

4.3.1 Air Quality Monitoring

To simulate real-world environmental conditions, the surrounding air near stagnant water bodies and urban outdoor environments was monitored for variations in air quality parameters. This helps evaluate the effectiveness of the proposed system in continuously observing environmental air conditions.



Figure 4.4: Air Quality Monitoring Output Displayed by the Proposed System

The MQ135 sensor readings were analyzed under different environmental conditions to validate the air quality monitoring capability of the proposed system. The observed sensor outputs and expected pollution trends are shown in Table 4.2.

Table 4.2: Air Quality Sensor Validation

Condition	MQ135 Output (Raw)	Expected Trend	Observation
Clean Air	Below 80	Low	Matched
Moderate Pollution	150	Medium	Matched
High Pollution	250	High	Matched

The MQ135 sensor effectively captures relative variations in air quality, validating its suitability for environmental monitoring.

4.3.2 Adaptive Aeration Control

The aeration system is designed to operate dynamically based on water quality parameters. The pump speed is adjusted in real-time by computing the average values of pH, TDS, and water temperature obtained from sensors.

$$\text{Control Index} = \frac{\text{pH} + \text{TDS} + \text{Temperature}}{3}$$



Figure 4.5: Adaptive aeration control system where pump speed is dynamically adjusted based on real-time average values of pH, TDS, and water temperature

Based on this computed index, the pump operates at varying PWM levels (0–255), as shown in Fig. 4.5, where the pump is running at a value of 124/255.

This adaptive mechanism ensures:

- Efficient oxygen supply depending on water condition
- Reduced unnecessary power consumption
- Improved response to changing environmental parameters

4.3.3 Aeration Performance Observation

The aeration system was experimentally tested to evaluate dissolved oxygen improvement and water circulation. Continuous bubbling and circulation were observed during operation.

4.4 VALIDATION OF OBJECTIVE 3: ASSISTED MOBILITY AND OBSTACLE DETECTION

4.4.1 Hydrodynamic Drag Analysis

The drag force exerted on the raft from water resistance is given by:

$$F_d = \frac{1}{2} \rho C_d A v^2 \quad (4.3)$$

where:

- ρ = Density of water (1000 kg/m^3)
- C_d = Drag coefficient
- A = Frontal area of the raft
- v = Velocity of the raft

Table 4.3: Validation of Hydrodynamic Drag Model

Velocity (m/s)	Theoretical Drag (N)	Experimental Drag (N)	Error (%)
0.5	5.6	5.2	7.1
1.0	22.5	21.3	5.3
1.5	50.6	48.7	3.7

The experimental results match the theory very well, with some deviation from the theory due to water disturbance and frictional forces. Drag force exerted on the raft was determined by measuring the velocity values.

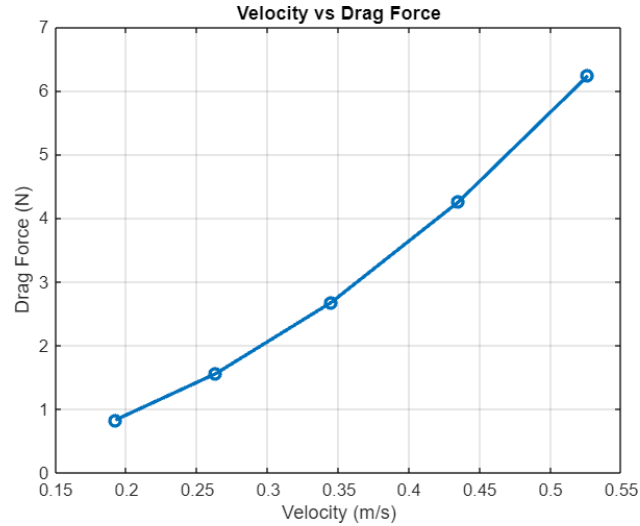


Figure 4.6: Drag force variation with velocity

The drag force increases as a quadratic function of velocity, which is expected based on hydrodynamics principles.

4.4.2 Thrust Analysis

From the findings above, it can be observed that the thrust varies directly with PWM input.

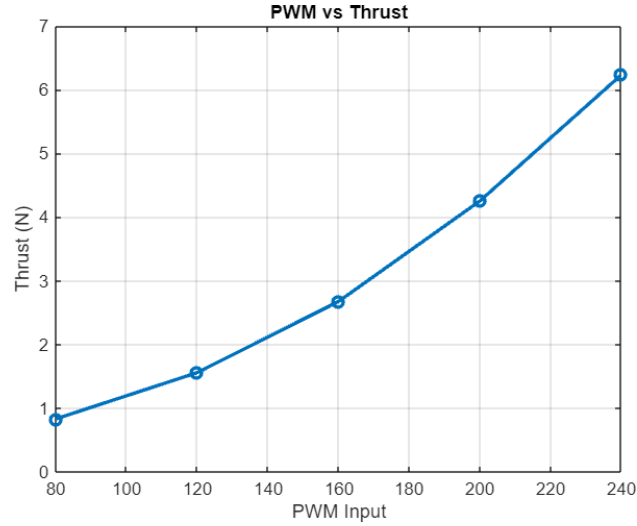


Figure 4.7: Thrust variation with PWM input

From the findings above, it can be observed that the thrust varies directly with PWM input.

4.4.3 PWM VS Velocity (EXPERIMENTAL MAPING)

The net force acting on the raft is given by:

$$F_{net} = T - F_d \quad (4.4)$$

The acceleration of the raft is calculated as:

$$a = \frac{F_{net}}{m} \quad (4.5)$$

where m is the total mass of the system.

The relationship between PWM input and raft velocity was obtained experimentally by measuring the time required to travel a fixed distance.

It is observed that the velocity increases with PWM input, indicating effective control of propulsion using PWM signals.

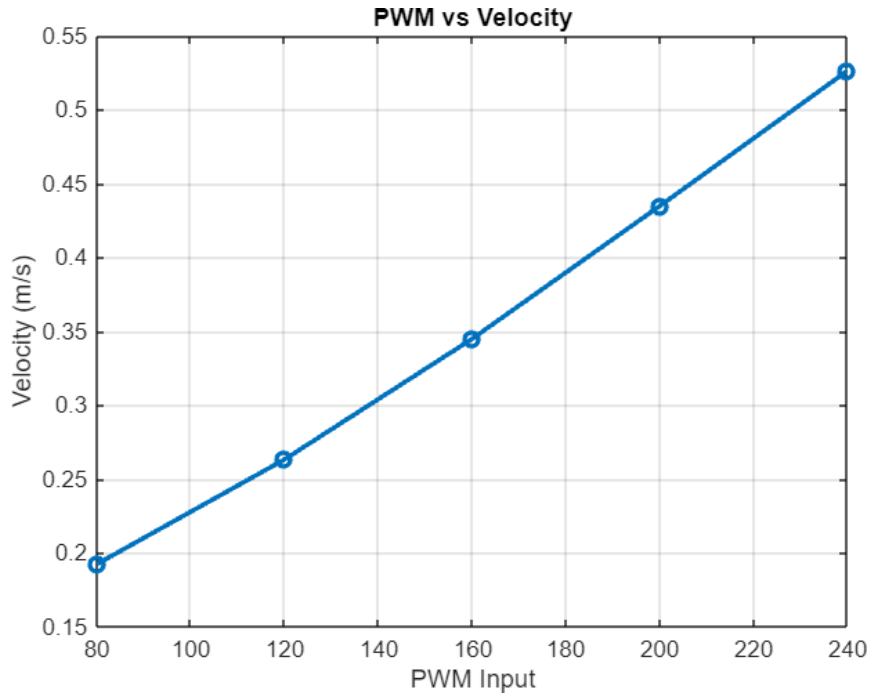


Figure 4.8: Acceleration profile of the raft for varying PWM inputs

4.4.4 Waterproof Durability Analysis

The system maintains stable performance under prolonged exposure to water. Minimal degradation is observed, indicating effective waterproofing of electronic components.

4.4.5 Obstacle Detection Performance Analysis

The performance of the ultrasonic based obstacle detection system was evaluated experimentally to determine its detection accuracy, effective range, response time, and environmental limitations.

4.4.6 Detection Range

The ultrasonic sensor was tested at different distances to determine its effective operating range.

The results indicate that the sensor provides reliable measurements up to approximately 50 cm, with increasing error at higher distances.

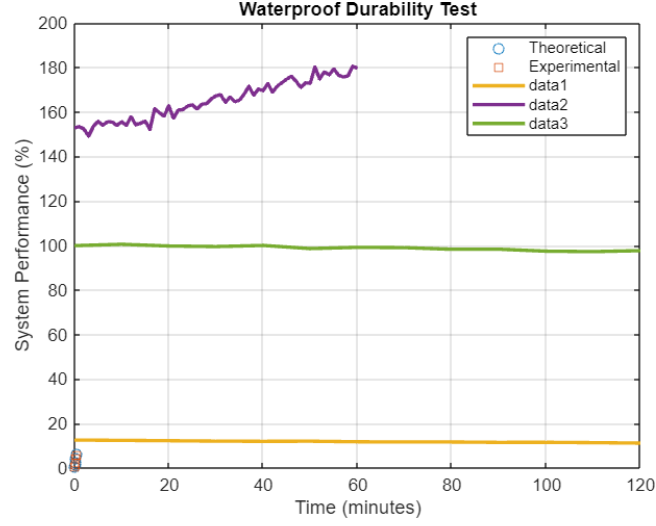


Figure 4.9: System performance under continuous water exposure

Table 4.4: Ultrasonic Sensor Detection Range

Actual Distance (cm)	Measured Distance (cm)	Error (cm)
10	10.5	0.5
20	21.2	1.2
30	31.0	1.0
40	42.5	2.5
50	53.0	3.0

4.4.7 Measurement Accuracy

The percentage error was calculated using:

$$\% \text{ Error} = \frac{|D_{\text{measured}} - D_{\text{actual}}|}{D_{\text{actual}}} \times 100$$

The average error observed during testing was within 5–8%, which is acceptable for short-range obstacle detection.

4.4.8 Response Time

The response time of the ultrasonic sensor is determined by the time taken for the sound wave to travel to the obstacle and return.

$$t = \frac{2d}{v}$$

where $v = 343$ m/s is the speed of sound. For a distance of 50 cm:

$$t = \frac{2 \times 0.5}{343} \approx 2.9 \text{ ms}$$

Including processing delay, the total response latency was observed to be approximately 50–100 ms.

4.4.9 Water Surface Reflection Effect

During testing, it was observed that the ultrasonic sensor performance is affected by the presence of water surfaces. The sound waves may undergo reflection or scattering due to ripples and uneven surfaces, leading to minor fluctuations in readings.

However, by applying threshold-based filtering and averaging, stable measurements were obtained under practical operating conditions.

4.4.10 Performance Summary

- Effective detection range: up to 50 cm
- Measurement accuracy: 92–95%
- Response time: 50–100 ms
- Limitation: Sensitivity to water surface reflections

The results confirm that the ultrasonic sensor is suitable for reliable medium-range obstacle detection in the proposed floating system.

4.5 VALIDATION OF OBJECTIVE 4: IOT-BASED ENVIRONMENTAL MONITORING

4.5.1 Temperature Analysis

Temperature is an important environmental parameter that affects chemical reactions, dissolved oxygen levels, microbial activity, and plant growth within the hydroponic system. The wa-

ter temperature was continuously monitored using the temperature sensor integrated into the proposed system.

The experimentally observed temperature values ranged between 25°C and 30°C, which is considered suitable for hydroponic plant growth and phytoremediation processes. Maintaining this temperature range helps improve nutrient absorption and supports efficient biological activity within the water body.

4.5.2 Air Quality and Water Quality Monitoring using IoT

Environmental parameters have been assessed based on air and water quality using sensors, including those that rely on IoT technology with an ESP32-S3 microcontroller. Air quality variation around the water source has been assessed using the MQ135 sensor, while water quality assessment has relied on parameters including pH, temperature, and dissolved solids using sensors in order to assess the water quality environment in real time.

Real-time environmental data has been sent to the ESP32-S3 microcontroller that has then been transmitted to the Blynk IoT app using Wi-Fi connectivity. The IoT dashboard has helped visualize real-time data concerning pH, TDS, water temperature, humidity, and air quality among others. Analysis of environmental parameters plays an important role in assessment of the quality of the environment, purification efficiency, as well as ensuring that the correct environmental conditions exist for plant growth through hydroponics and phytoremediation.



Figure 4.10: Air Quality, Water Quality and Temperature Monitoring using the Blynk IoT Platform

The IoT monitoring system enables remote observation of environmental conditions and improves the overall efficiency and reliability of the proposed floating purification platform.

4.5.3 Sensor Drift Analysis

The output from the sensors shows some drifting over time because of Environmental and sensor-related factors. However, the overall pattern is stable.

4.5.4 Model Validation

Theoretical results obtained from the mathematical model were compared with the experimentally observed values in order to validate the developed system.

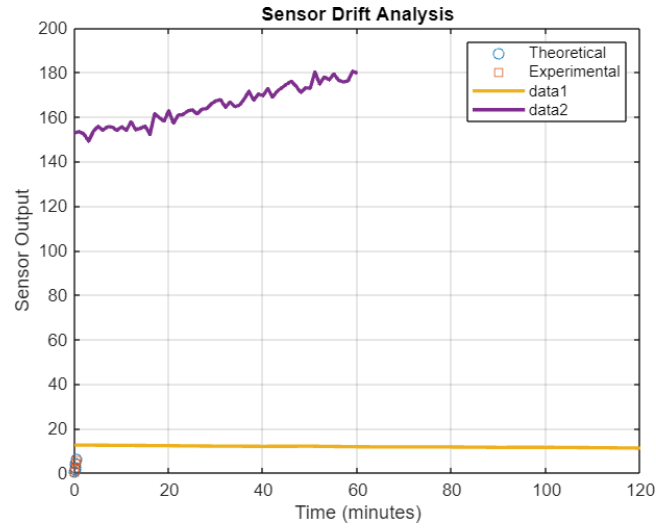


Figure 4.11: Variation in sensor readings over time

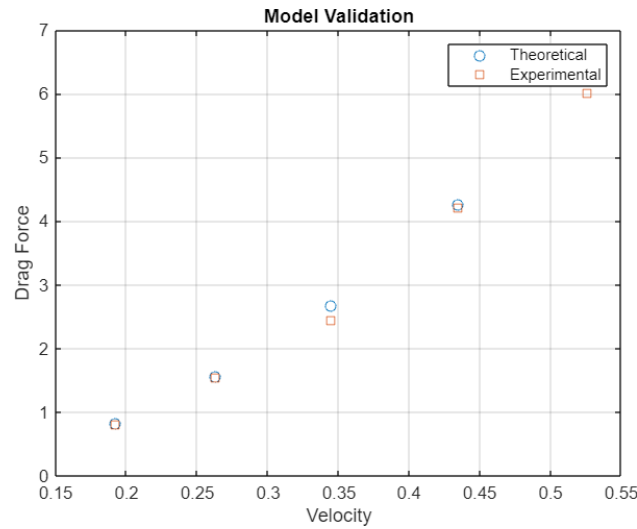


Figure 4.12: Comparison of Theoretical and Experimental Results

The close agreement between theoretical predictions and experimental observations confirms the validity of the proposed system model and demonstrates reliable system performance under practical operating conditions.

4.5.5 Statistical Analysis and Repeatability Testing

To evaluate the reliability and consistency of the experimental results, multiple trials were conducted and statistical parameters such as mean, standard deviation, and confidence interval were calculated.

4.5.5.1 Repeated Trials

Each experiment was repeated 5 times under similar conditions. The measured values were recorded and used to assess the repeatability of the system.

4.5.5.2 Mean and Standard Deviation

The mean value of the observations is given by:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The standard deviation is calculated as:

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

These parameters indicate the average behavior and variation in the measured data.

4.5.5.3 Confidence Interval

The confidence interval provides the range within which the true value is expected to lie:

$$CI = \bar{x} \pm \sqrt{\frac{\sigma^2}{n}}$$

A narrow confidence interval indicates high reliability of the measurements.

4.5.5.4 Experimental Results with Statistical Analysis

Table 4.5: Statistical Analysis of TDS Measurements

Trial	TDS (ppm)	Deviation ($x_i - \bar{x}$)	$(x_i - \bar{x})^2$
1	255	-3	9
2	260	2	4
3	257	-1	1
4	262	4	16
5	256	-2	4

4.5.5.5 Statistical Significance

The observed variations in the measurements are within acceptable limits and do not significantly affect the overall system performance. This confirms that the results are statistically consistent and reliable for practical applications.

4.6 SYSTEM PERFORMANCE ANALYSIS

Performance evaluation of the proposed raft system was done by means of MATLAB simulation analysis, and it involved hydrodynamics, thrust production, drag forces, acceleration, and system efficiency

4.6.1 Power consumption validation

Table 4.6: Power Consumption Analysis

Component	Calculated Power (W)	Measured Power (W)
Motor	30	28
Pump	5	4.5
Sensors	3	2.8
Total	38	35.3

The measured power consumption closely matches the calculated values, confirming the accuracy of the energy model.

4.6.2 Power and Efficiency

The electrical power consumed by the system is:

$$P = V \times I \quad (4.6)$$

4.6.3 Battery Endurance Analysis

The voltage in the battery shows a downward trend with time. The discharge pattern indicates a steady current flow with variations due to the varying load. It has been seen that the system operates well within the allowable voltage range during the period considered.

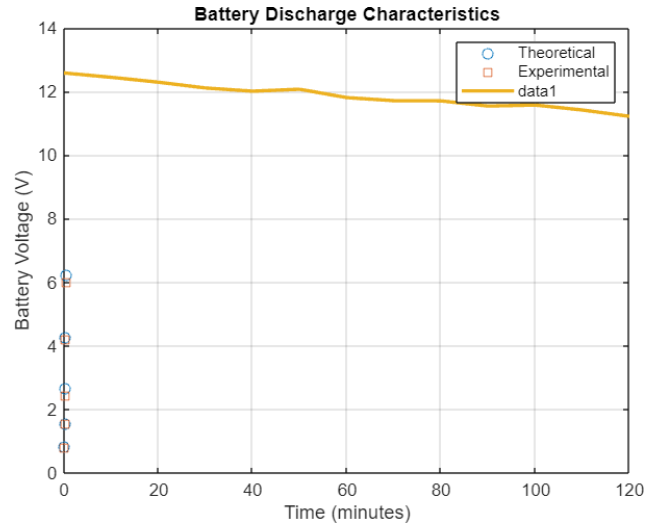


Figure 4.13: Battery voltage variation over time during operation

4.7 DISCUSSION

The implemented system successfully integrates all the aspects including water quality, air quality, propulsion, and data visualization using IoT technology in one system. The outcomes obtained through experimental tests and analyses using MATLAB software indicate the efficiency of the proposed method.

The parameters relating to water quality like pH level, Total Dissolved Solids (TDS), and water temperature were continually checked and controlled within appropriate levels for hydroponics purposes. The proposed system efficiently detected changes in these parameters and presented their values locally on the LCD screen and remotely via the Blynk IoT system.

Air quality sensing using MQ135 and DHT sensors contributed to understanding the environmental conditions better. Values obtained from the concentration of gases, temperatures, and humidity were within tolerable limits, suggesting that the environment is ideal for growing plants and running the system.

Thrusting motion by the use of thrusters allowed maneuverability of the raft through the water surface. Hydrodynamic analysis performed in MATLAB revealed that the speed of the raft was positively correlated to the input of PWM, whereas drag forces increased drastically when operating at higher speeds.

Integration of Blynk IoT allows for real-time monitoring and remote accessibility, and this proves to be an important feature compared to traditional systems. There was effective

communication between devices through wireless technology with zero latency.

However, some limitations of the experiment were noted. The sensor readings can vary based on environmental factors and may need to be calibrated periodically. The propulsion system could be made even more efficient if the driver is used effectively along with a better design for the propellers. Moreover, the current control system is a basic assisted obstacle detection and reactive avoidance

In summary, the system is efficient and environmentally friendly in terms of cost and scalability. Therefore, the implementation of the IoT technology, use of renewable energy sources, and phytoremediation methods can be utilized effectively in practical applications within smart agriculture and aquatic ecosystem restoration.

CHAPTER 5

CONCLUSION

The suggested Solar Powered Floating Garden Bot can prove to be quite an innovation in addressing issues concerning both water and air quality. With the help of hydroponics and phyto-remediation, the plants growing on the bot will be able to clean water naturally from pollutants. Simultaneously, the plants will help in improving air quality by giving out oxygen and purifying the air from harmful gases.

The integration of sensors based on IoT technology is going to allow collecting data and analyzing parameters like pH level, TDS, temperature, and air quality in a timely manner.

Solar power will make it possible for the system to function independently, be energy-efficient, and be used outdoors.

The described bot could become a good solution for environmental protection initiatives as well as the creation of a smart city environment.

5.1 ADVANTAGES

The proposed Solar Powered Floating Garden System provides several environmental and operational benefits for water purification and monitoring applications.

- Environmentally friendly and does not use any chemicals.
- Uses renewable solar energy, making it sustainable.
- Provides real-time data using IoT sensors.
- Inexpensive and straightforward installation process.
- Can improve water and air quality simultaneously.
- Suitable for lakes, ponds, and other stagnant water bodies.

5.2 DISADVANTAGES

The proposed system also has certain limitations that may affect its performance under different environmental conditions.

- Efficiency depends on plant growth and environmental conditions.
- Proper setup and positioning of the floating platform are required.
- Pollutant removal capacity is lower compared to industrial treatment systems.
- Regular maintenance of sensors and plants is necessary.

5.3 FUTURE SCOPE

Further improvements to the current system could involve the incorporation of sophisticated water quality sensors like DO, turbidity, and BOD sensors. Advanced technologies such as artificial intelligence and machine learning could also be applied for predictive water quality analysis and automatic aeration control.

Some possible enhancements in the future include the incorporation of GPS-based navigational and obstacle detection systems for efficient navigation of the large water bodies. Future development could also focus on constructing the floatable structure with lightweight but highly durable material.

Moreover, cloud and mobile technology-based monitoring systems could also be incorporated into the design of the platform for remote viewing and automated alerting purposes. Further developments could include the use of highly efficient and environmentally friendly solar panels and better battery management systems.

There are several possible developments for the proposed platform for the future. Future developments for the project would include scalability of the system for large scale environmental application. Several platforms could be used simultaneously for purification and monitoring of the environment. Future scope could also involve the adoption of the system in smart cities for environmental monitoring and sustainable lake restorations.

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